Exponent Rules

Name _____ **Date** _____ No calculator **Version 1**

KEY WORDS base · exponent · power · index · coefficient · reciprocal

A Multiplying powers ●○○

ANSWER ON THE SHEET

Remember: $a^m \times a^n = a^{m+n}$. Multiply the numbers in front as usual.

1. Simplify. Leave number answers as a power.

a) $x^3 \times x^4$ = _____ **b)** $a^2 \times a^7$ = _____ **c)** $2^5 \times 2^3$ = _____ **d)** $y \times y^6$ = _____

e) $3x^2 \times 4x^5$ = _____ **f)** $5a^3 \times 2a$ = _____ **g)** $x^2y \times x^3y^4$ = _____ **h)** $2ab^2 \times 6a^3b$ = _____

B Dividing powers ●○○

ANSWER ON THE SHEET

Remember: $a^m \div a^n = a^{m-n}$. Divide the numbers in front as usual.

2. Simplify. Leave number answers as a power.

a) $x^9 \div x^4$ = _____ **b)** $a^8 \div a$ = _____ **c)** $5^7 \div 5^3$ = _____ **d)** $12x^6 \div 3x^2$ = _____

e) $20a^5 \div 4a^4$ = _____ **f)** $x^5y^4 \div x^2y$ = _____ **g)** $\dfrac{18a^7}{6a^3}$ = _____ **h)** $\dfrac{8x^4y^6}{2xy^2}$ = _____

C Power of a power ●●○

ANSWER ON THE SHEET

Watch out: $(a^m)^n = a^{m\times n}$, and the number in front is raised to the power too: $(2a)^3 = 8a^3$.

3. Simplify. Leave number answers as a power.

a) $(x^3)^4$ = _____ **b)** $(a^5)^2$ = _____ **c)** $(2^3)^2$ = _____ **d)** $(y^2)^7$ = _____

e) $(2a^2)^3$ = _____ **f)** $(3x^4)^2$ = _____ **g)** $(x^2y^3)^4$ = _____ **h)** $(5ab^3)^2$ = _____

D Zero and negative powers ●●○

ANSWER ON THE SHEET

Remember: $a^0 = 1$ and $a^{-n} = \dfrac{1}{a^n}$

4. Work out the value.

a) 7^0 = _____ **b)** 2^{-3} = _____ **c)** 5^{-2} = _____ **d)** 10^{-1} = _____

e) $(-3)^2$ = _____ **f)** -3^2 = _____ **g)** $(-2)^3$ = _____ **h)** $4^0 + 4^1$ = _____

5. Write with positive exponents.

a) x^{-4} = _____ **b)** $3a^{-2}$ = _____ **c)** x^2y^{-3} = _____ **d)** $(2x)^{-2}$ = _____

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E Mixed practice ●●● WORK IN YOUR BOOK

6. Simplify.

a) $x^5 \times x^3 \div x^2$ **b)** $(a^3)^2 \times a^4$ **c)** $(2x^2)^3 \div 4x^5$

d) $12a^5b^3 \div 4a^2b$ **e)** $x^4 \times x^{-6}$ **f)** $(x^3y)^2 \times xy^3$

g) $3x^2 \times 2x^{-5}$ **h)** $\dfrac{a^7 \times a^2}{a^5}$ **i)** $(2a^2b)^3 \div 2ab$

7. Work out the value.

a) $2^3 \times 2^2$ **b)** $10^7 \div 10^4$ **c)** $(3^2)^2$ **d)** $5^6 \div 5^6$

e) $2^{-1} \times 2^4$ **f)** $(2 \times 5)^3$ **g)** $4^{-2} \times 4^3$ **h)** $6^2 \div 6^{-1}$

8. Find the value of n.

a) $x^n \times x^4 = x^{11}$ **b)** $(a^n)^3 = a^{12}$ **c)** $y^9 \div y^n = y^2$

d) $2^n = 16$ **e)** $5^n = 1$ **f)** $10^n = 0.01$

9. Each line has a mistake. Say what went wrong and write the correct answer.

a) $3x^2 \times 2x^3 = 6x^6$ **b)** $(2a^3)^2 = 2a^6$ **c)** $x^8 \div x^2 = x^4$

10. Write a simplified expression for each of these.

a) The area of a rectangle with length $4x^3$ and width $3x^2$.

b) The volume of a cube with sides of length $2a^2$.

c) The length of a rectangle with area $18x^7$ and width $6x^4$.

★ Extension FOR EARLY FINISHERS

11. Solve $2^x = 32$.

12. Simplify $\dfrac{(2a^3b)^2 \times 3ab^4}{6a^5b^3}$.

13. Which is larger, 2^{30} or 8^9? Show how you know.

14. Write $4^{-3} \times 8^2$ as a single power of 2.

15. A colony of bacteria doubles every hour. It starts with 2^3 cells. How many cells are there after 5 hours? Give your answer as a power of 2 and as a number.

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Exponent Rules

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS base · exponent · power · index · coefficient · reciprocal

A Multiplying powers ●○○

ANSWER ON THE SHEET

> **Remember:** $a^m \times a^n = a^{m+n}$. Multiply the numbers in front as usual.

1. Simplify. Leave number answers as a power.

a) $x^2 \times x^6$ = _____ **b)** $b^4 \times b^5$ = _____ **c)** $3^4 \times 3^2$ = _____ **d)** $m \times m^8$ = _____

e) $2x^3 \times 5x^4$ = _____ **f)** $4a^2 \times 3a$ = _____ **g)** $x^3y^2 \times xy^5$ = _____ **h)** $3a^2b \times 5ab^3$ = _____

B Dividing powers ●○○

ANSWER ON THE SHEET

> **Remember:** $a^m \div a^n = a^{m-n}$. Divide the numbers in front as usual.

2. Simplify. Leave number answers as a power.

a) $x^8 \div x^3$ = _____ **b)** $b^6 \div b$ = _____ **c)** $7^9 \div 7^4$ = _____ **d)** $15x^7 \div 5x^3$ = _____

e) $18a^6 \div 6a^5$ = _____ **f)** $x^6y^3 \div x^4y$ = _____ **g)** $\dfrac{24a^9}{8a^4}$ = _____ **h)** $\dfrac{10x^5y^7}{5x^2y^3}$ = _____

C Power of a power ●●○

ANSWER ON THE SHEET

> ⚠ **Watch out:** $(a^m)^n = a^{m\times n}$, and the number in front is raised to the power too: $(2a)^3 = 8a^3$.

3. Simplify. Leave number answers as a power.

a) $(x^2)^5$ = _____ **b)** $(b^4)^3$ = _____ **c)** $(3^2)^4$ = _____ **d)** $(m^3)^6$ = _____

e) $(3a^3)^2$ = _____ **f)** $(2x^5)^3$ = _____ **g)** $(x^3y^2)^3$ = _____ **h)** $(4a^2b)^2$ = _____

D Zero and negative powers ●●○

ANSWER ON THE SHEET

> **Remember:** $a^0 = 1$ and $a^{-n} = \dfrac{1}{a^n}$

4. Work out the value.

a) 9^0 = _____ **b)** 2^{-4} = _____ **c)** 3^{-2} = _____ **d)** 4^{-1} = _____

e) $(-4)^2$ = _____ **f)** -4^2 = _____ **g)** $(-3)^3$ = _____ **h)** $6^0 + 6^1$ = _____

5. Write with positive exponents.

a) a^{-5} = _____ **b)** $5x^{-3}$ = _____ **c)** $a^{-2}b^4$ = _____ **d)** $(3a)^{-2}$ = _____

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E Mixed practice ●●● WORK IN YOUR BOOK

6. Simplify.

a) $y^6 \times y^2 \div y^3$ **b)** $(b^2)^4 \times b^3$ **c)** $(3x^3)^2 \div 3x^4$

d) $20a^6b^4 \div 5a^2b^3$ **e)** $x^3 \times x^{-7}$ **f)** $(x^2y)^3 \times xy^2$

g) $4x^3 \times 2x^{-6}$ **h)** $\dfrac{b^8 \times b^3}{b^6}$ **i)** $(3ab^2)^2 \div 3ab$

7. Work out the value.

a) $3^2 \times 3^2$ **b)** $10^8 \div 10^6$ **c)** $(2^3)^2$ **d)** $7^4 \div 7^4$

e) $3^{-1} \times 3^3$ **f)** $(2 \times 10)^2$ **g)** $5^{-1} \times 5^3$ **h)** $2^5 \div 2^{-1}$

8. Find the value of n.

a) $x^n \times x^5 = x^{13}$ **b)** $(a^n)^4 = a^{20}$ **c)** $y^{10} \div y^n = y^4$

d) $3^n = 27$ **e)** $7^n = 1$ **f)** $10^n = 0.001$

9. Each line has a mistake. Say what went wrong and write the correct answer.

a) $4x^3 \times 2x^2 = 8x^6$ **b)** $(3a^2)^3 = 3a^6$ **c)** $y^{12} \div y^3 = y^4$

10. Write a simplified expression for each of these.

a) The area of a rectangle with length $5x^4$ and width $2x^3$.

b) The volume of a cube with sides of length $3a^3$.

c) The length of a rectangle with area $20x^9$ and width $4x^5$.

★ Extension FOR EARLY FINISHERS

11. Solve $3^x = 81$.

12. Simplify $\dfrac{(3a^2b)^2 \times 2a^3b}{6a^4b^2}$.

13. Which is larger, 3^{20} or 9^{11}? Show how you know.

14. Write $9^{-2} \times 27^2$ as a single power of 3.

15. A colony of bacteria doubles every hour. It starts with 2^4 cells. How many cells are there after 6 hours? Give your answer as a power of 2 and as a number.

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